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e-CAM82_CUOAGX



Product Datasheet

Revision 1.3

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1 Revision History

Rev	Date	Major Changes	Author
1.0	14-Jun-2022	Initial draft	Camera Products - Hardware Team
1.1	16-Nov-2022	Added Xavier support	Camera Products - Hardware Team
1.2	22-Feb-2023	4-Lane support	Camera Products - Hardware Team
1.3	14-Mar-2023	Added 2-Lane support	Camera Products - Hardware Team





2 Introduction

The e-CAM82_CUOAGX board is a camera product which is designed and developed by e-con Systems, a leading Embedded Product Design Services Company which specializes in the advanced camera solutions. This camera board targets the NVIDIA® Jetson AGX Orin™ / Xavier™ development kit.

e-CAM82_CUOAGX camera contains diagonal 12.86mm IMX485 sensor from STARVIS family of CMOS image sensor from Sony® and can be interfaced to the camera connector of the Jetson AGX Orin™ / Xavier™ using the e-CAM80_CUMI334_ADP and e-CAM30_HEXCUXVR_BASE_BRD board. e-con Systems provides the sample applications that demonstrates the features of this camera.

This document describes about the features of e-CAM84_CUMI485C_MOD board and the pin-outs of the connectors including the mechanical diagram.

3 Disclaimer

The specifications and features of e-CAM82_CUOAGX camera board are provided here as reference only and e-con Systems reserves the right to edit/modify this document without any prior intimation of whatsoever.

4 Description

Jetson AGX Orin™ / Xavier™ is a newly launched small size, low power, AI system-based evaluation boards developed by NVIDIA®. Jetson AGX Orin™ / Xavier™ development supports 4-lane MIPI CSI-2 camera connections. e-CAM82_CUOAGX uses these 4-lane MIPI CSI for connecting 8 MP camera modules.

e-CAM82_CUOAGX is a multi-board solution, which has three boards as follows:

- Camera module (e-CAM84_CUMI485C_MOD)
- Adaptor board (e-CAM80_CUMI334_ADP)
- Base board (e-CAM30_HEXCUXVR_BASE_BRD)

The camera module is a small, low-power, high performance 8 MP camera. This camera module is based on diagonal 12.86mm IMX485 sensor from STARVIS family of CMOS image sensor from SONY. The IMX485 is a diagonal 12.86mm optical form-factor, CMOS image sensor with variable speed shutter.

The following table lists the supported frame rates by e-CAM82_CUOAGX camera module.

Platform	Lans	Resolution	Frame Rate(fps) in 10-bit Output	Frame Rate(fps) in 12-bit Output
With Orin™/Xavier™	4-Lane	1920 x 1080*	90	90
		3840 x 2160	60	50
	2-Lane	1920 x 1080*	62	62
		3840 x 2160	39	33



Table 1: Supported Resolution and Frame Rates

*1944x1096 is cropped to 1920x1080 on the application end to skip the dummy bytes. 1944x1096 will be referred to as 1920x1080 throughout the document.

4.1 Features

The features of e-CAM82_CUOAGX are as follows:

- Multi-board solution
- 8 MP camera sensor
- Compatible with Jetson AGX Orin™ / Xavier™ development kits
- Light weight, versatile, and portable design.
- Imaging applications
 - 8 MP CMOS image sensor with diagonal 12.86mm optical form-factor
 - Field of View (FOV) angle is not the same for all preview resolutions
- Operating Voltage: 3.3V $\pm$ 5%
- RoHS compliant

5 Key Specifications

The following table lists the key specifications of e-CAM82_CUOAGX.

Description	Specification
Size (L x W)	30 mm × 30 mm
Video format	YUV(NV12)
Maximum image resolution	3840 x 2160 (8 MP)
Supported OS	Linux (Ubuntu)

Table 2: Key Specifications of e-CAM82_CUOAGX

5.1 CMOS Image Sensor Specifications

The following table lists the specifications of the CMOS image sensor used in this e-CAM82_CUOAGX camera board.

Sensor Specification	
Type / Optical size	1/1.2" CMOS Solid-State image sensor
Resolution	8 MP
Sensor Type	RGB RAW 10/12-bit
Pixel size	2.9 μ m x 2.9 μ m
Total number of Pixels	3864(H) x 2200(V)
Sensor Effective Area	3864(H) x 2180(V)

Table 3: CMOS Image Sensor Specification

For more information about the IMX485 sensor Datasheet, please contact SONY corporation.



6 Pin Description

e-CAM80_CUMI334_ADP adaptor board has two connectors CN1, CN2. The CN1 is dual row 26-pin Samtec connector, used for direct mating with the camera module, whereas CN2 is a single row 30-pin connector, used for connecting with Jetson AGX Orin™ / Xavier™ development kit through the IPEX cable. The dual row connector and IPEX connectors are Pin 1 to 1 mating type. You must note the given pin numbers and direction with respect to the adaptor board.

The pin descriptions of connectors are explained in the following sections.

6.1 Pin-out Details of Adaptor Board Dual Row Connector (CN1)

The following table lists the pin-out details of CN1 connector.

CN1 Pin No	Signal Name	Pin Type	Description
1	MIPI_CLK_N	OUTPUT	MIPI Clock Lane Differential Pair -
2	MIPI_DATA0_N	OUTPUT	MIPI Data Lane 0 Differential Pair -
3	MIPI_CLK_P	OUTPUT	MIPI Clock Lane Differential Pair +
4	MIPI_DATA0_P	OUTPUT	MIPI Data Lane 0 Differential Pair +
5	GND	POWER	Ground
6	GND	POWER	Ground
7	MIPI_DATA2_N	-	-
8	CSI_I2C_SCL	Bi-Directional	I2C Clock
9	MIPI_DATA2_P	-	-
10	CSL_I2C_SDA	Bi-Directional	I2C Data
11	GND	POWER	Ground
12	RESET	Input	Reset
13	MIPI_DATA3_N	-	-
14	INTERRUPT	Input	Sensor Power Down
15	MIPI_DATA3_P	-	-
16	GND	POWER	Ground
17	GND	POWER	Ground
18	CAM_HSYNC	Input	HSYNC
19	MIPI_DATA1_N	OUTPUT	MIPI Data Lane 1 Differential Pair -
20	CAM_VSYNC	Input	VSYNC
21	MIPI_DATA1_P	OUTPUT	MIPI Data Lane 1 Differential Pair +
22	XMASTER	-	-
23	GND	POWER	Ground
24	TRIGGER	-	-
25	CAM_FLASH	-	-
26	VCC_3P3	POWER	3.3V Supply

Table 4: Adaptor Board CN1 Connector Pin Description Details



6.2 Pin-out Details of Adaptor Board IPEX Connector (CN2)

The following table lists the pin-out details of CN2 connector.

CN3 Pin No	Signal Name	Pin Type	Description
1	VCC_3P3	POWER	3.3V Power supply for camera board
2	VCC_3P3	POWER	3.3V Power supply for camera board
3	VCC_1P8	POWER	1.8V Power supply for camera board
4	GND	POWER	Ground
5	GND	POWER	Ground
6	INTERRUPT	OUTPUT	
7	CSI_I2C_SCL	Bi-Directional	I2C Clock
8	CSI_I2C_SDA	Bi-Directional	I2C Data
9	GND	POWER	Ground
10	MIPI_DATA2_N	OUTPUT	MIPI Data Lane 0 Differential Pair -
11	MIPI_DATA2_P	OUTPUT	MIPI Data Lane 0 Differential Pair +
12	TRIGGER	INPUT	Trigger pin
13	HSYNC	Bi-Directional	Hsync pin
14	GND	POWER	Ground
15	MIPI_DATA1_N	OUTPUT	MIPI Data Lane 1 Differential Pair -
16	MIPI_DATA1_P	OUTPUT	MIPI Data Lane 1 Differential Pair +
17	GND	POWER	Ground
18	GND	POWER	Ground
19	MIPI_DATA0_N	OUTPUT	MIPI Data Lane 0 Differential Pair -
20	MIPI_DATA0_P	OUTPUT	MIPI Data Lane 0 Differential Pair +
21	RESET	INPUT	Reset pin
22	GND	POWER	Ground
23	XMASTER	INPUT	Xmaster pin
24	MIPI_CLK_N	OUTPUT	MIPI Data Lane clock Differential Pair -
25	MIPI_CLK_P	OUTPUT	MIPI Data Lane clock Differential Pair +
26	GND	POWER	Ground
27	MIPI_DATA3_N	OUTPUT	MIPI Data Lane 0 Differential Pair -
28	MIPI_DATA3_P	OUTPUT	MIPI Data Lane 0 Differential Pair +
29	CAM_FLASH	Bi-Directional	Cam flash pin
30	VSYN	Bi-Directional	Vsync pin
31-49	GND	POWER	Ground



Table 5: Adaptor Board IPEX CN2 Connector Pin Description Details

6.3 Connector Part Numbers

The following table lists the connectors and cables used in e-CAM82_CUOAGX and its compatible mating connectors.

Connector	Description	Manufacturer	Part Number
Mating connector	0.80 mm, Board - Board, 26 pin	Samtec	ERF8-013-05.0-L-DV-L-K-TR
e-CAM80_CUMI334_ADP IPEX connector (CN2) for connecting with Jetson AGX Orin™ / Xavier™ development kit through IPEX cable	CONN IPEX 30 Pos Right angle SMT	TE Connectivity	20682-030E-02
IPEX cable used for connecting e-CAM80_CUMI334_ADP with Jetson AGX Orin™ development kit	30 pin IPEX cable R	Würth Electronics	81214-530B-300-1

Table 6: e-CAM84_CUMI485C_MOD Connector Details

7 Electrical Specification

The following sections list the electrical specification, recommended operating conditions and power consumption details of e-CAM82_CUOAGX.

The values described in this section are measured in e-con Systems lab and this can be used as reference only. The current measurements are typical values and are subject to change for different camera boards under different conditions. However, these values can be taken as a reference for power estimation and power supply design.

7.1 Recommended Operating Condition

The following table lists the recommended operating condition of e-CAM84_CUMI485C_MOD.

Platform	Lanes	Resolution	Frame Rate(fps) in 10-bit Output	Frame Rate(fps) in 12-bit Output	Power Consumption (W) in 10-bit Output	Power Consumption (W) in 12-bit Output
With Orin™ / Xavier™	4-lane	1920 x 1080	90	90	-	
		3840 x 2160	60	50	-	
With Orin™ / Xavier™	2-lane	1920 x 1080	62	62	-	
		3840 x 2160	39	33	-	

Table 7: Recommended Operating Condition

e-CAM82_CUOAGX does not requires any power sequence since it required only 3.3V power supply for operation.

7.2 Functional Temperature Range

The functional temperature range of e-CAM82_CUOAGX is listed in following table.

Temperature Range	Parameter Description
-30°C to 70°C	Electrically functional operating range

Table 8: Functional Temperature Range

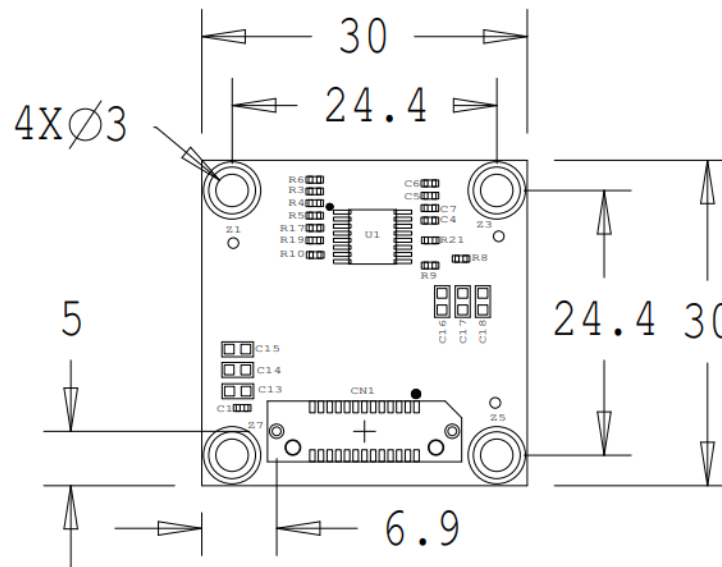
Notes:

- As temperature increases, the noise level also increases.
- Default lens (optional) supplied with this camera has operating temperature range of -20°C to 60°C. You can choose wider operating temperature lens as per your requirements.
- Performance guaranteed temperature range -10°C to 60°C.

8 Mechanical Specification

The adaptor board and camera board of e-CAM82_CUOAGX are 30 mm x 30 mm in dimension (outline tolerance of $\pm 0.2\text{mm}$).

The front and rear views of the e-CAM82_CUOAGX adaptor board and camera board with its dimensions are shown in the following figures.



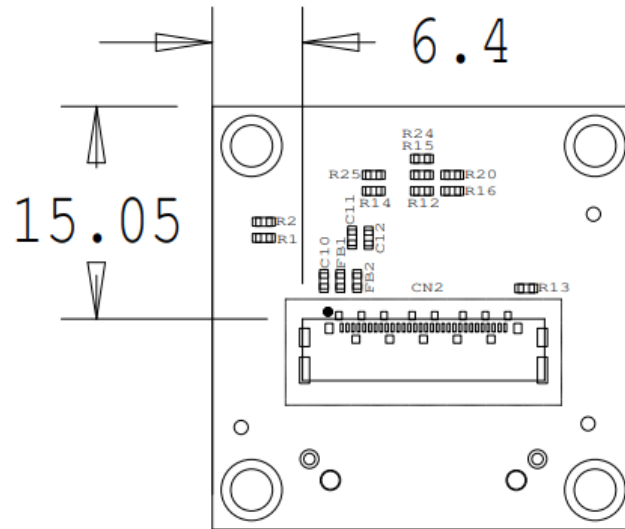


Figure 2: Rear View of e-CAM82_CUOAGX Adaptor Board Mechanical Dimensions

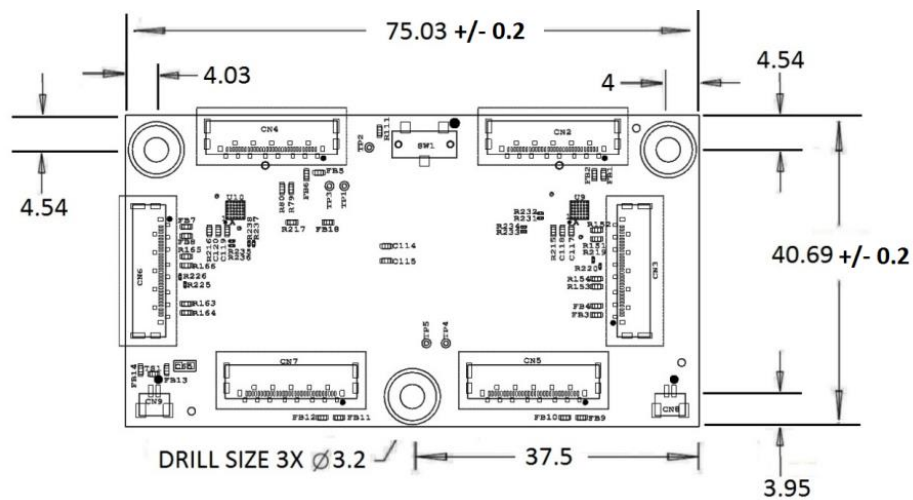


Figure 3: Front View of e-CAM30A_HEXCUTX2 Base Board Mechanical Dimensions



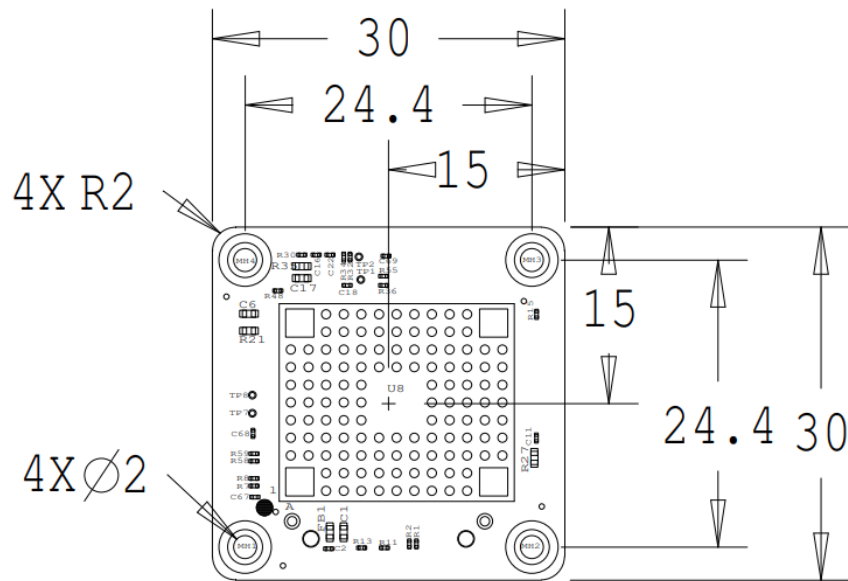


Figure 4: Top View of e-CAM84_CUMI485C_MOD_H02R1 Mechanical Drawing

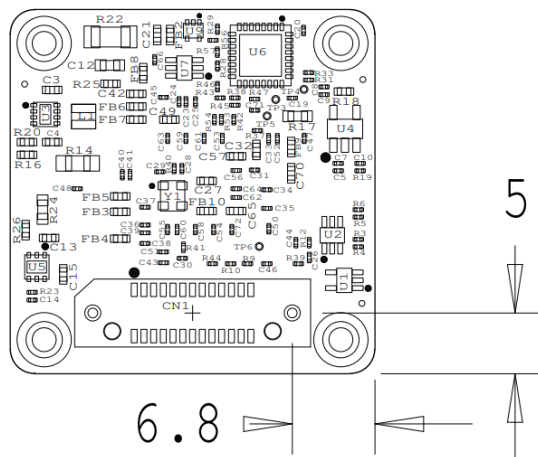


Figure 5: Bottom View of e-CAM84_CUMI485C_MOD_H02R1 Mechanical Drawing (Mirrored)

Note:

- All dimensions are in mm.
- Board outline dimension tolerance should be within $\pm 0.2\text{mm}$.
- Further mechanical details will be provided based on your request.

For e-CAM84_CUMI485C_MOD module board mechanical dimension information, please refer to the e-CAM84_CUMI485C_MOD Datasheet.



Support

Contact Us

If you need any support on e-CAM82_CUOAGX product, please contact us using the Live Chat option available on our website - <https://www.e-consystems.com/>

Creating a Ticket

If you need to create a ticket for any type of issue, please visit the ticketing page on our website - <https://www.e-consystems.com/create-ticket.asp>

RMA

To know about our Return Material Authorization (RMA) policy, please visit the RMA Policy page on our website - <https://www.e-consystems.com/RMA-Policy.asp>

General Product Warranty Terms

To know about our General Product Warranty Terms, please visit the General Warranty Terms page on our website - <https://www.e-consystems.com/warranty.asp>

